

Himank Dave

steadyfall.github.io

✉ hddave@uwaterloo.ca
in himank-dave
🌐 steadyfall

🎓 Education

University of Waterloo

Bachelor of Mathematics, Computational Mathematics

> Coursework: Object-Oriented Programming, Data Structures and Algorithms, Data-Intensive Distributed Computing

Waterloo, Canada

Sep 2022 - Present

🧰 Experience

Software Developer Intern, Adaptive Traffic Control

Miovision

May 2026 - Aug 2026

Kitchener, ON

Software Developer Intern, Embedded Infrastructure

Geotab

Jan 2025 - Aug 2025

Oakville, ON

- > Scaled and managed fleet of self-hosted **GitLab Runners** on Google **Kubernetes** Engine (GKE) using **Helm**, increasing concurrency by **50x** and enabling **autoscaling** to eliminate 1h+ CI queues for thousands of jobs.
- > Built scalable **gRPC microservice** in **Golang** to retrieve build artifacts and repository files with **commit-aware caching**, reducing asset retrieval latency for Software-in-the-Loop (SIL) testing framework from **~20s** to **under 2s**.
- > Eliminated **security configuration drift** across **250+ repositories** by engineering a **Python-based CLI tool** that automates manifest synchronization, JSON schema validation and atomic rollbacks.
- > Reverse-engineered legacy Ubuntu 16 production **Docker images** into **Dockerfiles**, upgraded to Ubuntu 22 & **optimized image layers** to reduce size by **30%** for reproducible firmware builds.
- > Containerized **Windows-only** embedded development environment using Wine, automating **firmware builds** for **500k+ products** while ensuring legacy software compatibility.

Software Developer Intern

Cactus Creatives

May 2024 - Aug 2024

Remote

- > Built a self-hosted **uptime monitoring platform** using **Node.js**, **Socket.IO**, **Redis**, and **React**; deployed it on VPS via Docker and **detected 15+ downtime incidents** in the first month.
- > Developed an **ETL pipeline** using **Flask** and **Pandas** to aggregate 50k+ hierarchical records, while enabling interactive visualizations with **React** and **D3.js**.
- > Engineered domain-specific multilingual chatbot leveraging **OpenAI API**, achieving **85%** accuracy on proprietary dataset; integrated into client workflows.

Python Developer Intern

Cactus Creatives

May 2023 - Aug 2023

Remote

- > Developed realtime content management system in **Django** to display analytics for waste management company.
- > Migrated 50k+ records from **MySQL** to **PostgreSQL** via custom Python scripts, eliminating data integrity issues and reducing query times by **40%**.
- > Implemented **CI/CD workflows** for 15+ repositories using **GitHub Actions**, cutting deployment times by **40%**.

🔧 Projects

Open Source Contributor - Sure 🌐 | Ruby, Docker, Github Actions

- > Improved container release infrastructure for an open-source (**8K+ stars**) finance app by reducing Docker image build time from **35min** to **6min** through **multi-architecture matrix builds** and **registry-backed caching**.
- > Modernized Sure's contributor experience by upgrading **devcontainer tooling** with Git-aware shell prompts, **Ruby LSP support** for VS Code, Git autocomplete, Codespaces compatibility, and worker-service configuration fixes.

fspeek 🌐 | Golang, SQLite, Bubbletea

- > Built TUI for **HTTP directory browsing** with **byte-range metadata extraction**, eliminating full downloads and reducing latency across large media directories.
- > Implemented **SQLite cache-first architecture** (24h TTL) with pluggable Nginx/HTML parsers, cutting redundant HTTP calls and improving load performance.
- > Engineered **concurrent prefetch** system (semaphore + 100ms debounce) with nonce-based guards; enabling fast, racefree navigation without full downloads.

deris 🌐 | Golang

- > Built minimal **key-value store** with network socket interface supporting CRUD operations & high throughput.
- > Achieved **50k+ operations/sec** and **<1ms latency** using Go concurrency primitives and buffered I/O.
- > Implemented **Append-Only File snapshots** with **SHA256 checksums** to ensure data integrity and enable reliable recovery on shutdown.

⚙️ Skills

Languages Python, Golang, TypeScript, JavaScript, C++, Bash

Frameworks Django, gRPC, React, Next.js, Node.js, Socket.IO, TailwindCSS

Tools Docker, Kubernetes, Helm, Terraform, GitLab CI, GitHub Actions, Google Cloud Platform (GCP)